

A Comparison of Cause-specific Melanoma Mortality and All-cause Mortality in Survival Analyses after Radiation Treatment for Uveal Melanoma

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Objective: To determine the causes and patterns of mortality after uveal melanoma radiation.

Design: A cohort study from a single institution was performed. Mortality was modeled using semiparametric survival techniques. All cause and cause-specific mortality analyses were performed. Mortality was compared with expected mortality from the U.S. census data.

Participants: A total of 731 patients were studied, and 710 (97%) of these had medium or large melanomas. The mean tumor diameter was 11.3 mm, and the mean tumor thickness was 5.8 mm. Ciliary body was involved in 122 (17%) of patients. Complete follow-up was available on 99.6% (728 of 731) of patients.

Main Outcome Measures: The authors analyzed the distribution and causes of post-treatment mortality.

Results: The 5- and 10-year all-cause Kaplan-Meier survival rates were 75.6% and 62.3%, respectively. Both melanoma risk factors (older age, ciliary body involvement, and larger tumor diameter) and nonmelanoma risk factors (older age and medical condition) were significant prognostic factors of all-cause mortality. Deaths from nonmelanoma causes accounted for 91 (42.3%) of 215 deaths. The 5-year and 10-year estimates of nonmelanoma deaths were 8.3% and 15.9%, respectively. Nonmelanoma mortality was similar to that observed in the general U.S. population (91 observed, 98.1 expected). Melanoma metastases accounted for 124 (57.7%) of 215 deaths. The 5- and 10-year estimates for probability of metastatic death were 16.1% and 21.8%, respectively. The largest tumor diameter was the best predictor for melanoma mortality; ciliary body involvement, older age, and distance from the fovea also were significant in multivariate analyses.

Conclusion: A significant proportion of patients with uveal melanoma die of nonmelanoma causes after radiation. In analyzing prognostic factors, considerable information may be lost if analyses are based on all-cause mortality rather than cause-specific mortality. *Ophthalmology* 1998;105:2035-2045

Numerous publications have examined mortality after enucleation for uveal melanoma.¹ However, melanoma-related and all-cause mortality after brachytherapy and charged particle radiation are not as well characterized. Using meta-analyses, the 5-year all-cause mortality rates of medium-sized tumors have been estimated to be approximately 30%.¹ Actuarial postradiation melanoma-related mortality rates have usually been lower.²⁻⁸

The published cause-specific mortality rates after various uveal melanoma therapies range from 16% to 51%, with 5-year cause-specific melanoma mortality ranging from 14% to 38%.⁹⁻²¹ The competing-cause nonmelanoma mortality rates range from 6% to 32%.¹¹⁻²¹ In general, mela-

noma mortality was higher in studies with longer follow-up, after enucleation, and in patients treated in prior eras.¹¹⁻²¹ The reliability of these estimates is difficult to ascertain. In some studies, selection biases were present and patients were excluded from the analysis if they died of other causes before the minimum follow-up time.¹⁸

At diagnosis, patients with uveal melanoma tend to be older than for many other malignancies and, thus, are at higher risk of death from other causes. For example, in a cohort of individuals that match our radiated cohort for age, gender, and race, the expected percent of individuals dying of any cause within 5 and 10 years is 9.9% (standard error [SE] = 1.0%) and 21.7% (SE = 1.4%), respectively. The older age of patients with uveal melanoma complicates both the calculation and interpretation of tumor-related mortality rates. Some treated patients with uveal melanoma die of nontumor-related causes without sufficient follow-up to ascertain either local control or distant disease status, and a number of patients with treated uveal melanoma will die of other causes before existing metastatic melanoma is detected.

Conventional techniques for modeling tumor-related mortality usually treat patients who die of nontumor-related

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